

FOST TRAINING RULE OF THE ROAD TEST NOV 16

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(G) Vol I and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1 hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

PART 1

- a) To whom and where do these rules apply? (R. 1a) (1)
 - b) Nothing in the Rules shall interfere with the operation of any special rules made by the Government of any State with respects toWhat? (R. 1c) (2)
- a) Name the Sections of Part B-“Steering and sailing rules”? (3)
 - b) Define the following words or terms:
 - i) ‘Sailing Vessel’ (R. 3c) (1)
 - ii) ‘Seaplane’ (R. 3e) (1)
 - iii) ‘Restricted Visibility’ (R. 3l) (1)

3. Name the six factors that, when considering Safe Speed, shall be among those taken into account by vessels with operational radar. (R. 6b) **(3)**

4. a) How should a vessel determine if risk of collision exists? (R. 7a) **(1)**

b) Assumptions shall not be made on the basis of ... What? (R. 7c) **(1)**

5. Rule 8 is 'Action to Avoid Collision'. Part 'c' states that if there is sufficient sea room, alteration of course alone maybe the most effective action to avoid a close quarters situation provided that it is.... What? (R. 8c) **(3)**

6. Complete the following;

A vessel which, by ____ __ ____, is required not to ____ the passage or ____ of another vessel ____, when required by the circumstance of the case, take ____ action to allow ____ for the safe passage of the other vessel. (R. 8fi) **(3)**

7. What shall a vessel do when using a Traffic Separation Scheme adopted by the organization? (R. 10bi, ii, iii) **(3)**

8. a) Where one of two vessels is to keep out of the way of another the other shall.... What? (R. 17ai) **(2)**
- b) When may a stand-on vessel take action to avoid collision by her manoeuvre alone? (R. 17aii) **(1)**
- c) If a power driven vessel takes action to avoid collision with another power driven vessel what shall, if the circumstance of the case admit, she not do? (R. 17c) **(1)**
9. What should a vessel engaged in fishing when underway, so far as possible, keep out of the way of? (R. 18c) **(1)**
10. a) In which Part, and in what Section is Rule 19? **(1)**
- b) How should a vessel proceed with regards to Restricted Visibility? (R. 19b) **(1)**
- c) If, when another vessel has been detected by radar alone, a close quarters situation is developing and / or risk of collision exists what avoiding action shall, so far as possible, be avoided? (R. 19di, ii) **(2)**

PART 2

You are OOW1 on HMS Richmond. You are on passage from Plymouth to Portland and en-route will be conducting replenishment with an RFA tanker.

11. According to the WPP, the day of your arrival into Portland a visiting SSN is due to sail. The CO asks you whether or not the SSN must comply with the provisions of the Rules with regards to the display of navigation lights. What do the Rules state? (R. 1e)

(3)

12. The CO also notice at the WPP that HMS OCEAN will be conducting flight operations during ship's RAS phase and he ask you: who is the give way vessel? (ATP1(d))

(1)

13. The port lookout interrupts you to report a group of yachts ahead. He reports three yachts from R10 to R30. The second yacht is on a steady bearing.

a) Who is the give way vessel? (R. 18a)

(1)

b) What action should you take and what should the result of that action be? (R. 16 & R. 8c)

(2)

c) The last yacht is displaying a cone apex down. What does this indicate? (R. 25e)

(1)

14. Now clear of the yachts the sky looks ominously black and visibility begins to reduce in the gloom.

a) Can you display your navigation lights before sunset? (R. 20c)

(1)

b) What is the minimum range of your masthead light? (R. 22a) (1)

c) What is the arc of visibility of your stern light? (R. 21c) (1)

15. Rain begins to fall reducing visibility to 6nm. At G10 you make out the following lights and assess that the vessel will be passing clear down you starboard side.

		W	W
		R	W
		W	W
		R	
G	G		

a) What do these lights indicate? (R. 24a) (2)

b) What is the day signal for this vessel? (R. 24av & 24e) (2)

c) In restricted visibility what sound signal and at what interval would this vessel sound if both vessels were manned? (R. 35c & 35e) (2)

As the passage is proceeding unhindered and, surprisingly, without the constraint of restricted visibility you decide to quiz your YO on his RoR knowledge.

16. a) i) If main propulsion was lost what type of vessel would you become? (R. 3f)

(1)

ii) What lights would you display by night and; (R. 27a)

(1)

iii) If in restricted visibility, what sound signal would you make? (R. 35c)

(1)

b) If you stopped making way though the water what difference, if any, would this make to the lights you would display? (R. 27a)

(1)

17. Should the ME's not be able to conduct repairs in time and you ran aground what would the day, night and sound signal be in this instance? (R. 30a, 30d, 35g & 35h)

(2)

18. What is the minimum diameter of a black ball and what should be the minimum vertical spacing between shapes? (Annex I, 6ai & 6b)

(1)

19. Should the worst occur and you needed to indicate distress or need of assistance name 6 of the 15 signals you could display / sound in accordance with the Rules. (Annex IV 1.)

(3)

Unfortunately the starboard lookout interrupts your quiz to report that one of previously reported sailing vessels have altered course and is now closing at G160.

20. a) What type of situation is this and who is give way? (R. 13a, b)

(1)

b) The yacht maintains its course and speed and you determine that a close quarters situation is developing. What sound signal shall you make? (R. 34d)

(1)

c) The yacht carrying on despite repeated sounding of the signal in Q18b. When may you take action to avoid collision by your manoeuvre alone? (R. 17 aii)

(1)

Part 2 _____ /30

TOTAL _____ /62

PART 1, ANSWERS

1. a) To whom and where do these rules apply? (R. 1a) (1)

These Rules shall apply to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels.

- b) Nothing in the Rules shall interfere with the operation of any special rules made by the Government of any State with respects toWhat? (R. 1c) (2)

Nothing in these Rules shall interfere with the operation of any special rules made by the Government of any State with respect to additional station or signal lights, shapes or whistle signals for ships of war and vessels proceeding under convoy, or with respect to additional station or signal lights, or shapes for fishing vessels Engaged in Fishing as a fleet. (2)

2. a) Name the Sections of Part B-“Steering and sailing rules”? (3)

Section I: Conduct of vessels in any condition of visibility

Section II: Conduct of vessels in sight of one another

Section III: Conduct of vessels in restricted visibility

- b) Define the following words or terms:

iv) ‘Sailing Vessel’ (R. 3c) (1)
The term ‘sailing vessel’ means any vessel under sail provided that propelling machinery, if fitted, is not being used.

v) ‘Seaplane’ (R. 3e) (1)
The word ‘seaplane’ includes any aircraft designed to manoeuvre on the water.

vi) ‘Restricted Visibility’ (R. 3l) (1)
The term ‘Restricted Visibility’ means any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms or any other similar causes.

3. Name the six factors that, when considering Safe Speed, shall be among those taken into account by vessels with operational radar. (R. 6b) (3)

(i) the characteristics, efficiency and limitations of the radar equipment;
(ii) any constraints imposed by the radar range scale in use;
(iii) the effect on radar detection of the sea state, weather and other sources of interference;
(iv) the possibility that small vessels, ice and other floating objects, may not be detected by radar at an adequate range;
(v) the number, location and movement of vessels detected by radar;
(vi) the more exact assessment of the visibility that may be possible when radar is used to determine the range of vessels or other objects in the vicinity.

4. a) How should a vessel determine if risk of collision exists? (R. 7a)
(1)

Every vessel shall use all available means appropriate to the prevailing circumstances and conditions to determine if Risk of Collision exists. If there is any doubt such risk shall be deemed to exist.

- b) Assumptions shall not be made on the basis of ... What? (R. 7c)
(1)

Assumptions shall not be made on the basis of scanty information, especially scanty radar information.

5. Rule 8 is 'Action to Avoid Collision'. Part 'c' states that if there is sufficient sea room, alteration of course alone maybe the most effective action to avoid a close quarters situation provided that it is.... What? (R. 8c)
(3)

If there is sufficient sea room, alteration of course alone may be the most effective action to avoid a Close-Quarters situation provided that it is made in good time, is substantial and does not result in another Close-Quarters situation.

6. Complete the following; (3)

A vessel which, by any of these Rules, is required not to Impede the passage or safe passage of another vessel shall, when required by the circumstance of the case, take early action to allow sufficient sea room for the safe passage of the other vessel. (R. 8fi)

7. What shall a vessel do when using a Traffic Separation Scheme adopted by the organization? (R. 10bi, ii, iii)
(3)

(i) proceed in the appropriate Traffic Lane in the general direction of traffic flow for that Lane;
(ii) so far as practicable keep clear of a traffic Separation Line or Separation Zone;
(iii) normally join or leave a Traffic Lane at the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable.

8. a) Where one of two vessels is to keep out of the way of another the other shall.... What? (R. 17ai) (1)

Where one of two vessels is to keep out of the way the other shall keep her course and speed.

- b) When may a stand-on vessel take action to avoid collision by her manoeuvre alone? (R. 17a) (1)

The latter vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

- c) If a power driven vessel takes action to avoid collision with another power driven vessel what shall, if the circumstance of the case admit, she not do? (R. 17c) (1)

A power-driven vessel which takes action in a Crossing situation in accordance with subparagraph (a)(ii) of this Rule to avoid collision with another power-driven vessel shall, if the circumstances of the case admit, not alter course to port for a vessel on her own port side.

9. What should a vessel engaged in fishing when underway, so far as possible, keep out of the way of? (R. 18c) (1)

A vessel Engaged in Fishing when underway shall, so far as possible, keep out of the way of:

- (i) a vessel *Not Under Command*;
(ii) a vessel *Restricted in her Ability to Manoeuvre*.

10. a) In which Part, and in what Section is Rule 19? (1)

(Part B) Section III - Conduct of Vessels in Restricted Visibility

- b) How should a vessel proceed with regards to Restricted Visibility? (R. 19b) (1)

Every vessel shall proceed at a Safe Speed adapted to the prevailing circumstances and conditions of Restricted Visibility. A power-driven vessel shall have her engines ready for immediate manoeuvre.

- c) If, when another vessel has been detected by radar alone, a close quarters situation is developing and / or risk of collision exists what avoiding action shall, so far as possible, be avoided? (R. 19d, ii) (2)

- (i) *an alteration of course to port for a vessel forward of the beam, other than for a vessel being Overtaken;*
(ii) *an alteration of course towards a vessel abeam or abaft the beam.* (2)

PART 2, ANSWERS

You are OOW1 on HMS Richmond. You are on passage from Plymouth to Portland and en-route will be conducting replenishment with an RFA tanker.

11. According to the WPP, the day of your arrival into Portland a visiting SSN is due to sail. The CO asks you whether or not the SSN must comply with the provisions of the Rules with regards to the display of navigation lights. What do the Rules state? (R. 1e)

(3)

Whenever the Government concerned shall have determined that a vessel of special construction or purpose cannot comply fully with the provisions of any of these Rules with respect to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signalling appliances, such vessel shall comply with such other provisions in regard to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound signalling appliances, as her Government shall have determined to be the closest possible compliance with these Rules in respect of that vessel.

12. The CO also notice at the WPP that HMS OCEAN will be conducting flight operations during ship's RAS phase and he ask you: who is the give way vessel? (ATP1(d))

(1)

OCEAN is the give way vessel. Ships engaged in RAS (other than Vertrep) have the right of way over aircraft carriers and other ships engaged in flight operations.

13. The port lookout interrupts you to report a group of yachts ahead. He reports three yachts from R10 to R30. The second yacht is on a steady bearing.

a) Who is the give way vessel? (R. 18a)

(1)

You are; A power-driven vessel underway shall keep out of the way of: (iv) a sailing vessel.

b) What action should you take and what should the result of that action be? (R. 16 & R. 8c)

(2)

R. 16; Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.

R. 8c; If there is sufficient sea room, alteration of course alone may be the most effective action to avoid a Close-Quarters situation provided that it is made in good time, is substantial and does not result in another Close-Quarters situation.

- c) The last yacht is displaying a cone apex down. What does this indicate? (R. 25e)

(1)

A vessel proceeding under sail when also being propelled by machinery shall exhibit forward where it can best be seen a conical shape, apex downwards.

14. Now clear of the yachts the sky looks ominously black and visibility begins to reduce in the gloom.

- a) Can you display your navigation lights before sunset? (R. 20c)

(1)

Yes; The lights prescribed by these Rules shall, if carried, also be exhibited from sunrise to sunset in Restricted Visibility and may be exhibited in all other circumstances when it is deemed necessary

- b) What is the minimum range of your masthead light? (R. 22a)

(1)

In vessels of 50 metres or more in length:

- a masthead light, 6 miles;
- a sidelight, 3 miles;
- a sternlight, 3 miles;
- a Towing light, 3 miles;
- a white, red, green or yellow all-round light, 3 miles.

- c) What is the arc of visibility of your stern light? (R. 21c)

(1)

'Sternlight' means a white light placed as nearly as practicable at the stern showing an unbroken light over an arc of the horizon of 135 degrees and so fixed as to show the light 67.5 degrees from right aft on each side of the vessel.

15. Rain begins to fall reducing visibility to 6nm. At G10 you make out the following lights and assess that the vessel will be passing clear down you starboard side.

		W	W
		R	W
		W	W
		R	
G	G		

- a) What do these lights indicate? (R. 24a)

(2)

A power-driven vessel, possibly over 50m in length, seen from the starboard side, restricted in her ability to manoeuvre engaged in towing operations with a length of tow exceeding 200m.

- b) What is the day signal for this vessel? (R. 24av & 24e) (2)

A diamond shape where it can best be seen on the towing vessel and on the tow.

- c) In restricted visibility what sound signal and at what interval would this vessel sound if both vessels were manned? (R. 35c & 35e) (2)

R35c;a vessel Restricted in her Ability to Manoeuvre.... shall, instead of the signals prescribed in paragraphs (a) or (b) of this Rule, sound at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts.

A vessel Towed or if more than one vessel is Towed the last vessel of the Tow, if manned, shall at intervals of not more than 2 minutes sound four blasts in succession, namely one prolonged followed by three short blasts. When practicable, this signal shall be made immediately after the signal made by the Towing vessel.

As the passage is proceeding unhindered and, surprisingly, without the constraint of restricted visibility you decide to quiz your YO on his RoR knowledge.

16. a) i) If main propulsion was lost what type of vessel would you become? (R. 3f) (1)

A vessel Not Under Command (NUC)

- ii) What lights would you display by night and; (R. 27a) (1)

(a) A vessel Not Under Command shall exhibit:

- (i) two all-round red lights in a vertical line where they can best be seen;***
(iii) when making way through the water.. sidelights and a sternlight.

NB. NO masthead light

- iii) If in restricted visibility, what sound signal would you make? (R. 35c) (1)

At intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short blasts.

b) If you stopped making way through the water what difference, if any, would this make to the lights you would display? (R. 27a) (1)

If you were stopped and making no way through the water you would only display two all round red lights in a vertical line.

17. Should the ME's not be able to conduct repairs in time and you ran aground what would the day, night and sound signal be in this instance? (R. 30a, 30d, 35g & 35h) (2)

R. 30 a&d; Anchor lights & two all-round red lights in a vertical line where they can best be seen.

R. 35 g&h; At intervals of not more than one minute give three separate and distinct strokes on the bell immediately before and after a rapid ringing of the bell for about 5 seconds in the forepart of the vessel and immediately after the ringing of the bell the gong shall be sounded rapidly for about 5 seconds in the after part of the vessel

A vessel aground may in addition sound an appropriate whistle signal.

18. What is the minimum diameter of a black ball and what should be the minimum vertical spacing between shapes? (Annex I, 6ai & 6b) (1)

- ***a ball shall have a diameter of not less than 0.6 metre;***
- ***The vertical distance between shapes shall be at least 1.5 metres.***

19. Should the worst occur and you needed to indicate distress or need of assistance name 5 of the 15 signals you could display / sound in accordance with the Rules. (Annex IV 1.) (3)

- (a) a gun or other explosive signal fired at intervals of about a minute;***
- (b) a continuous sounding with any fog-signalling apparatus;***
- (c) rockets or shells, throwing red stars fired one at a time at short intervals;***
- (d) a signal made by radiotelegraphy or by any other signalling method consisting of the group ...---...(SOS) in the Morse Code;***
- (e) a signal sent by radiotelephony consisting of the spoken word 'Mayday';***
- (f) the International Code Signal of distress indicated by NC;***
- (g) a signal consisting of a square flag having above or below it a ball or anything resembling a ball;***

- (h) flames on the vessel (as from a burning tar barrel, oil barrel, etc.);**
- (i) a rocket parachute flare or a hand flare showing a red light;**
- (j) a smoke signal giving off orange-coloured smoke;**
- (k) slowly and repeatedly raising and lowering arms outstretched to each side;**
- (l) the radiotelegraph alarm signal;**
- (m) the radiotelephone alarm signal;**
- (n) signals transmitted by emergency position-indicating radio beacons;**
- (o) approved signals transmitted by radio-communication systems, including survival craft radar transponders.**

Unfortunately the starboard lookout interrupts your quiz to report that one of previously reported sailing vessels have altered course and is now closing at G160.

20. a) What type of situation is this and who is give way? (R. 13a, b) (1)

This is an Overtaking Situation.

A vessel shall be deemed to be Overtaking when coming up with another vessel from, a direction more than 22.5 degrees abaft her beam, that is, in such a position with reference to the vessel she is Overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.

- b) The yacht maintains its course and speed and you determine that a close quarters situation is developing. What sound signal shall you make? (R. 34d) (1)

When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by a light signal of at least five short and rapid flashes.

- c) The yacht carrying on despite repeated sounding of the signal in Q18b. When may you take action to avoid collision by your manoeuvre alone? (R. 17 aii) (1)

.....may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.